



NPTEL Course on “Computer System Optimization for ML”

Lecture #15



Agenda

- **Recap**
- **DNN accelerators**
 - DianNao [1]
 - DaDianNao [2]
 - ShiDianNao [3]
 - **Eyeriss [4]**

[1] Chen, Tianshi, et al. "Diannao: A small-footprint high-throughput accelerator for ubiquitous machine-learning." ACM SIGARCH Computer Architecture News 42.1 (2014): 269-284.

[2] Chen, Yunji, et al. "Dadiannao: A machine-learning supercomputer." *2014 47th Annual IEEE/ACM International Symposium on Microarchitecture*. IEEE, 2014.

[3] Du, Zidong, et al. "ShiDianNao: Shifting vision processing closer to the sensor." *Proceedings of the 42nd Annual International Symposium on Computer Architecture*. 2015.

[4] Chen, Yu-Hsin, et al. "Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks." *IEEE journal of solid-state circuits* 52.1 (2016): 127-138.

Accelerator Architecture in DianNao

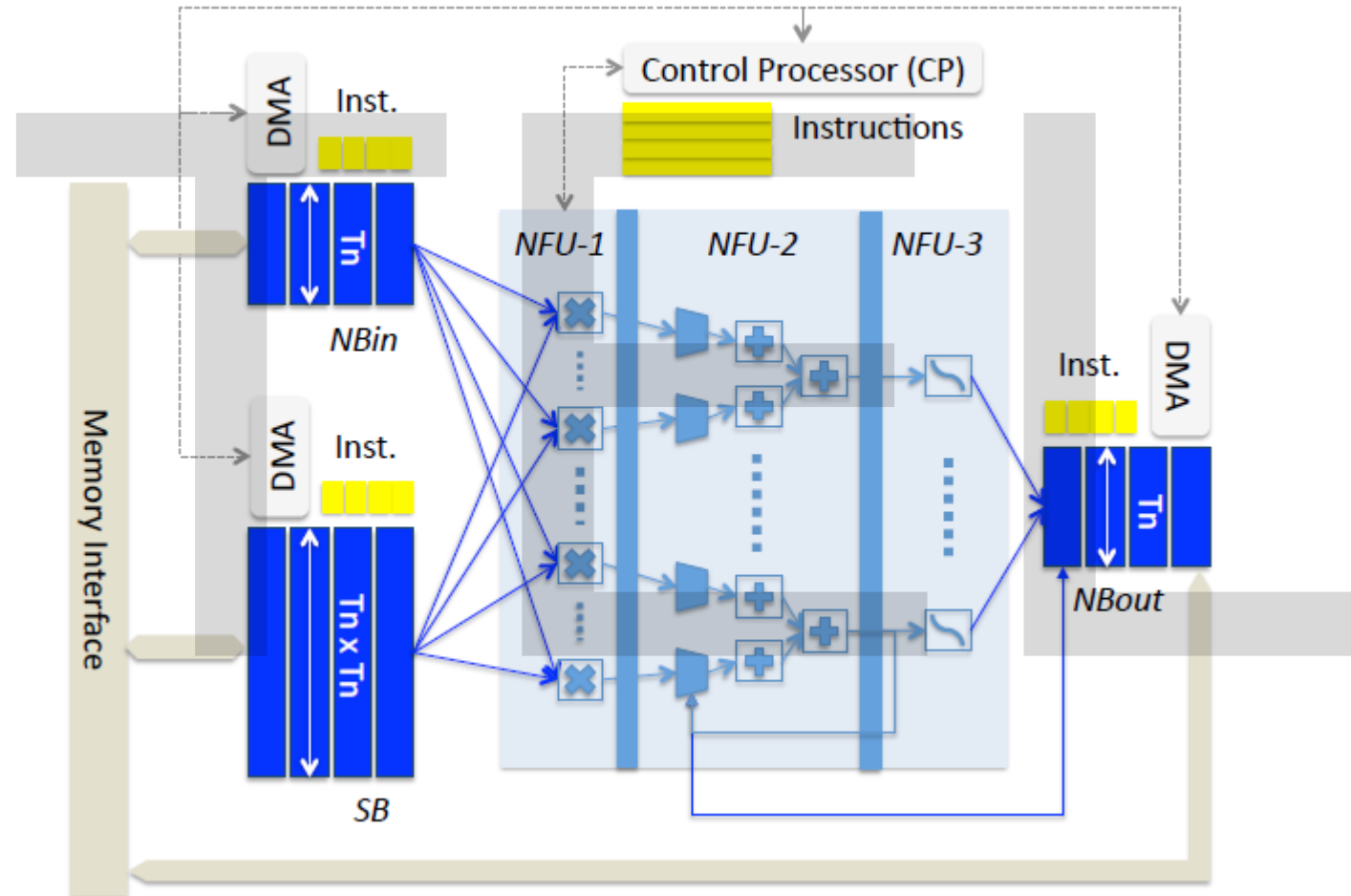
■ Storage

- NBin: To store input activations
- SB: To store synapses
- NBout: To store outputs

■ Neural functional unit (NFU)

- NFU-1: multiplier
- NFU-2: adder tree
- NFU-3: non linear

■ Control instructions



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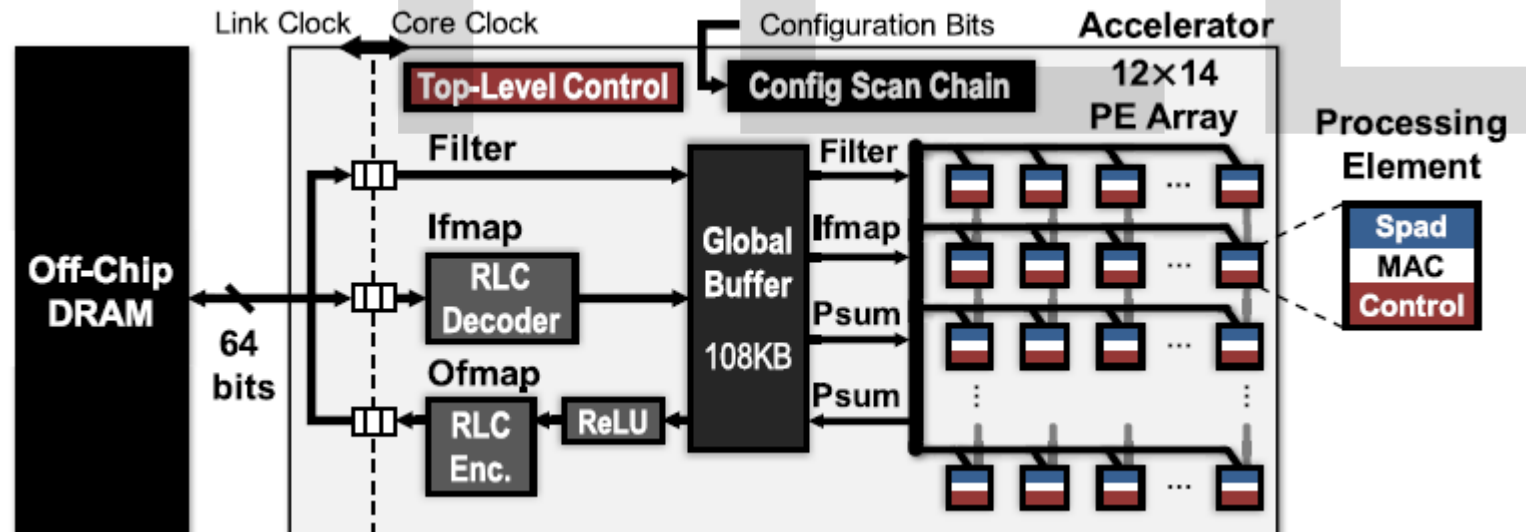
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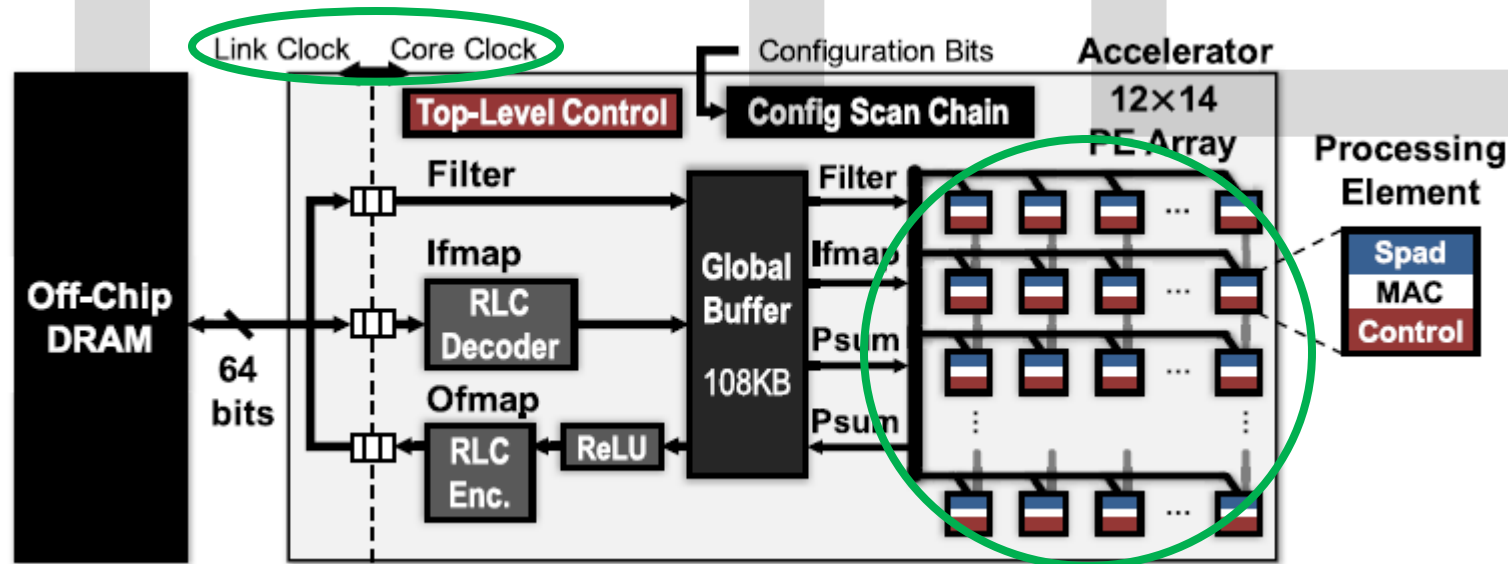
[4] Chen, Yu-Hsin, et al. "Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks." IEEE journal of solid-state circuits 52.1 (2016): 127-138.

- **Eyeriss: An Energy-efficient Reconfigurable Accelerator for Deep Convolutional Neural Networks**
- **What is new?**
 - New dataflow
 - New architecture
 - Encoding and decoding technique
 - Fabricated chip (a co-processor)



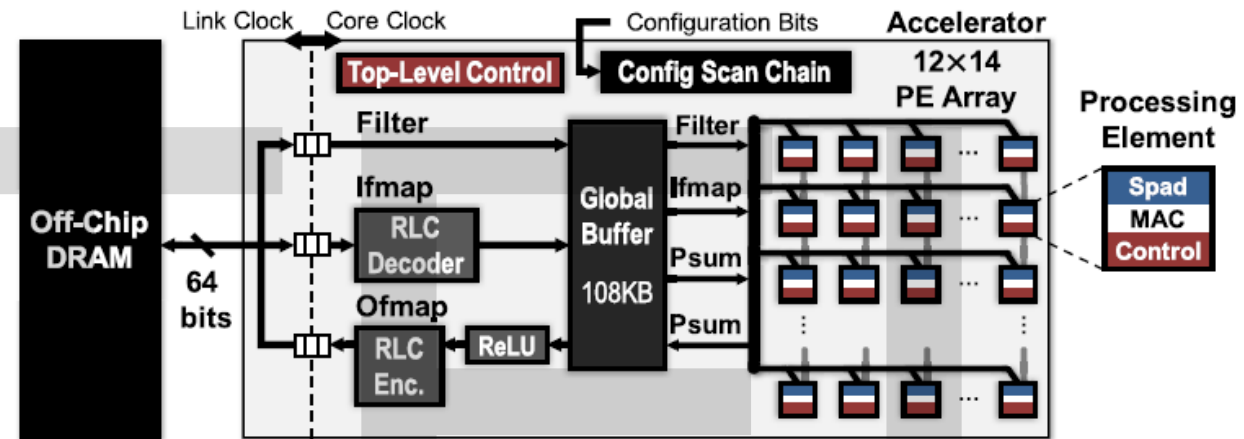
Overview of the Architecture (1)

- **Two clock domains**
 - Core clock domain for processing
 - Link clock domain for communication with DRAM
- **Communication**
 - PE and Global Buffer (GLB)
 - PE and PE
 - PE and scratchpad (spad)



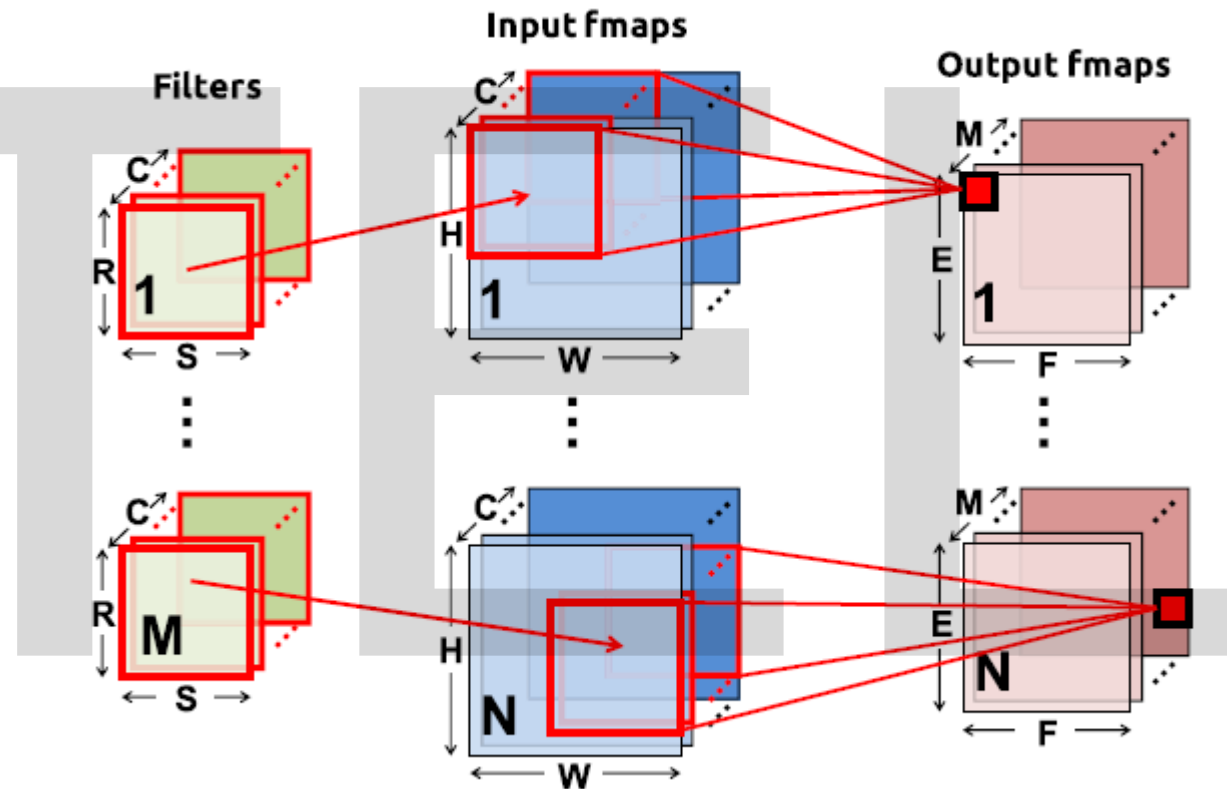
Overview of the Architecture (2)

- **Two levels of control**
- **Top level**
 - Traffic between DRAM and GLB
 - Traffic between GLB and PE array (NoC)
 - Operation of RLC and ReLU
- **Low level**
 - Control logic in each PE
- **Note: PEs execute independent of each other**
 - Each PE starts processing whenever fmaps(?) or psums(?) arrives
 - Not like systolic array



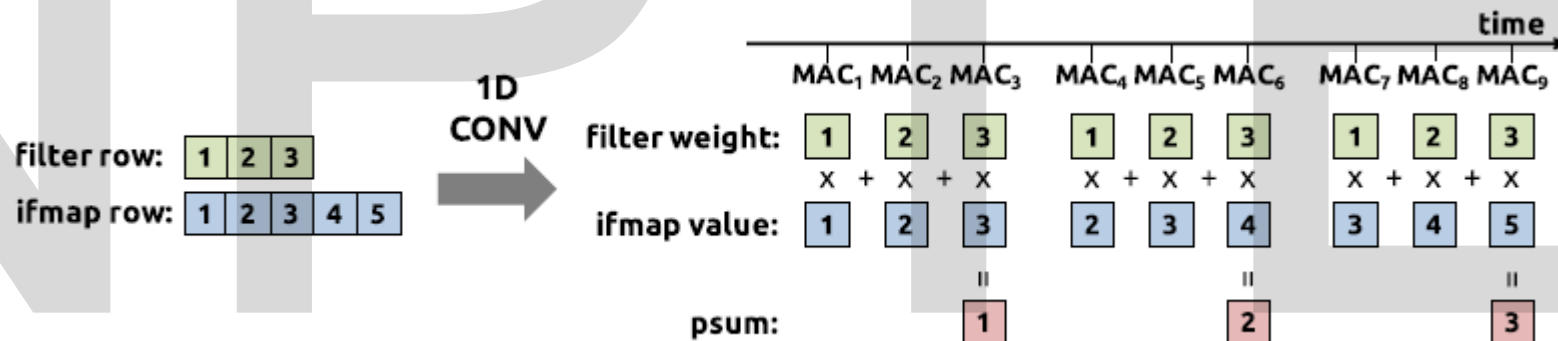
Data Reuse

- **How many times a filter is reused?**
 - $E \times F$ (in the same ifmap plane)
- **How many times an ifmap pixel is reused?**
 - $R \times S$ (in the same filter plane)
- **Each filter weight is reused across the batch of N if maps**
- **Each ifmap is reused across M filters**
- **Idea: $C \times R \times S$ psum values must be computed as soon as possible**
 - To minimize storage and data movement



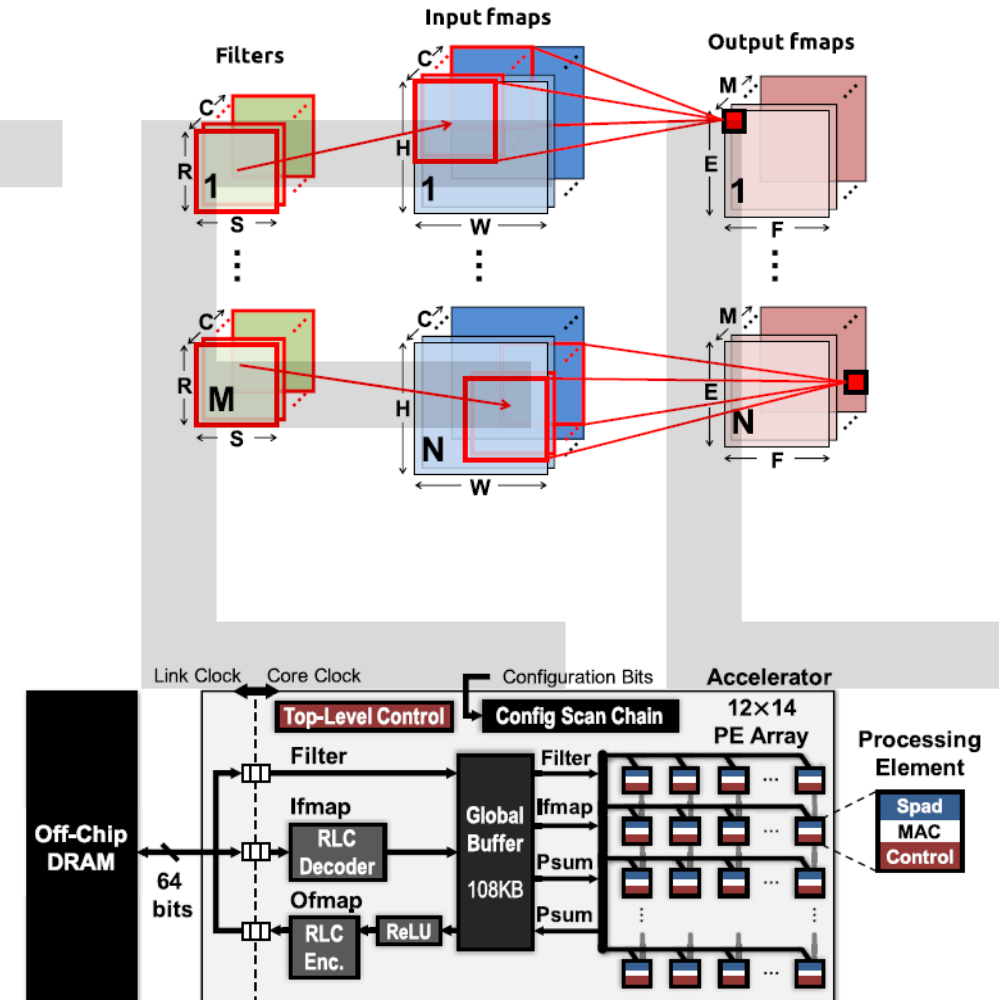
Row Stationary Dataflow (1)

- Each primitive operates on one row of filter weights and one row of ifmap to produce one row of psums
 - psums are accumulated to generate ofmap value
- Example with 1-D convolution

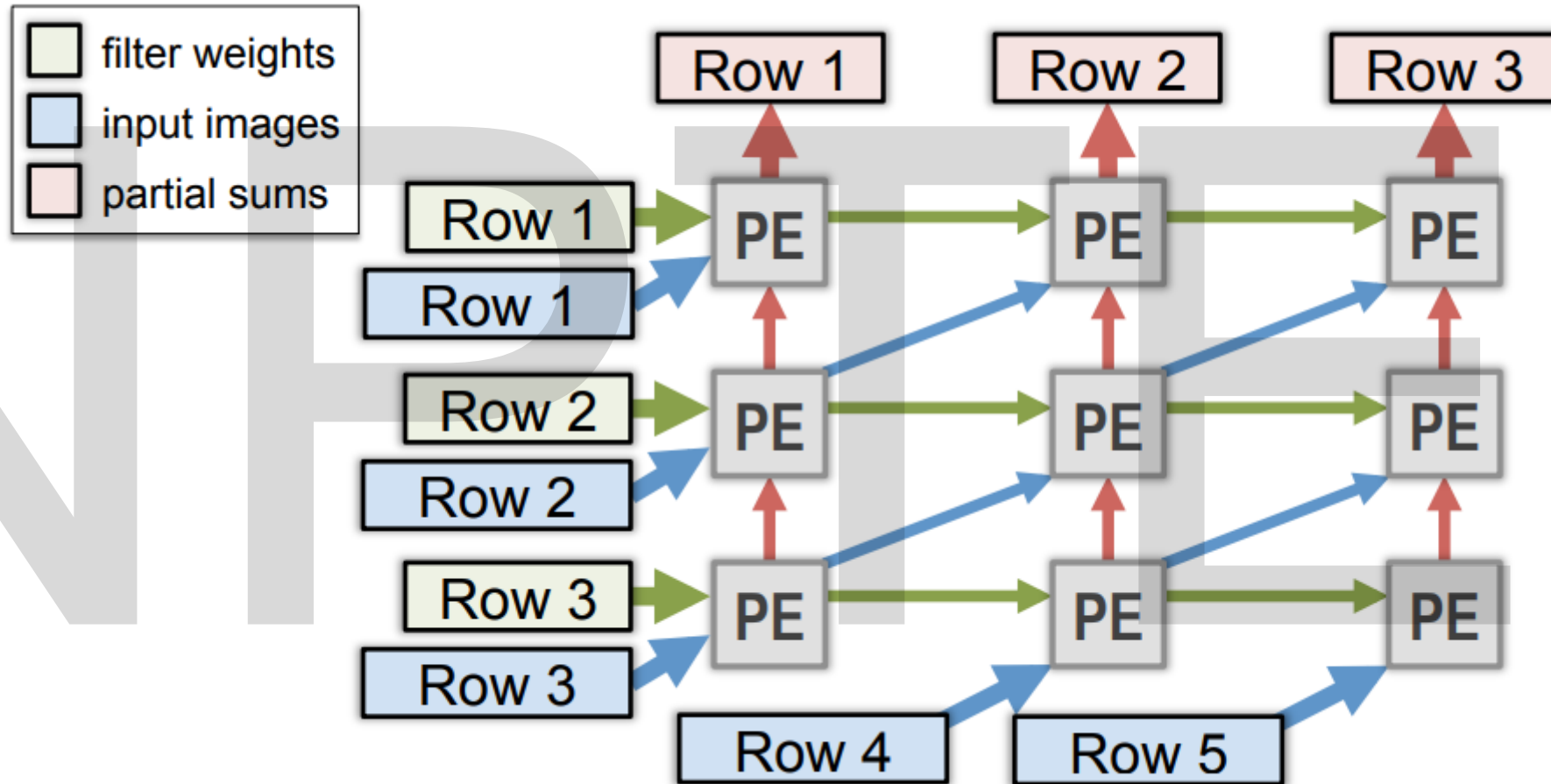


Row Stationary Dataflow (2)

- How many psums need to be stored?
 - Same as filter row size (S)
- Where should we store?
 - spad
- spad capacity
 - Weights (S), ifmaps (S), psum (1)
 - $2S+1$

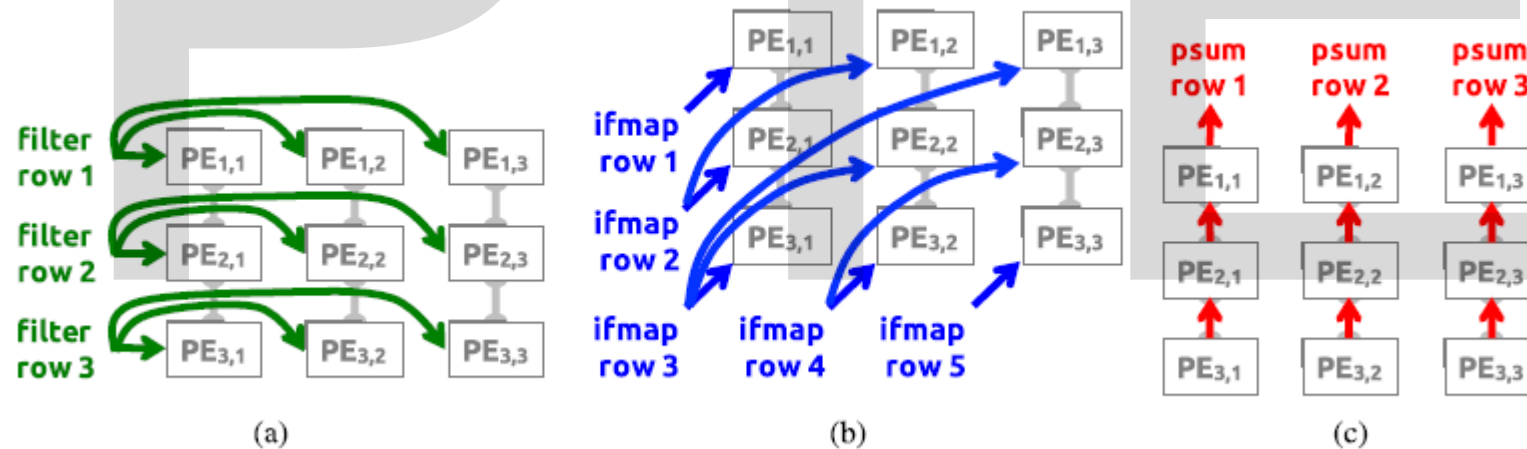


Row Stationary Dataflow (3)



Row Stationary Dataflow (4)

- **Extension to 2-D convolution**
 - Consists of several 1-D convolution
- **A PE array size of 14×12 is incorporated in the paper**





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)

